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Portable routing system and method of use

Abstract

A portable router table that includes a top surface of a bench that is adapted to support at least one workpiece. The portable router table also includes a bottom surface of the bench that is spaced apart from the top surface. The portable router table also includes at least one bottom slot defined in the bottom surface of the bench. The bench is adapted to engage with a bench clamp inside of the at least one bottom slot for clamping the portable router table with a support structure. In one instance, the bench may be cantilevered from the support structure and the bottom surface of the bench is engaged with the support structure.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure is directed to a portable router table for woodworking operations.

BACKGROUND ART

(2) Portable routers and similar portable cutting devices are versatile woodworking tools that are commonly used to cut various types of workpiece or stock. Generally, portable routers are used to cut various types of features and/or designs into a workpiece as dictated by the woodworking project based on a router bit equipped to the router. While a woodworker may use a portable router to cut various types of features and designs into a workpiece, it is difficult and demanding to cut precisely and accurately depending on the size, shape, or configuration of the workpiece.

(3) To address these difficulties and problems, a woodworker may select a router table or similar apparatus to help stabilize and guide the portable router during a cutting operation. While such assisting apparatuses are provided in the market and useful in controlling a portable router, woodworkers that use such apparatuses may have to succumb to a more compact woodworking environment or purchase additional apparatuses to prevent such compact environment. In one example, portable router tables require a fair amount of workspace in a woodworking environment given the footprint of such router tables. While portable router tables are also commonly used in

woodworking projects, such portable router tables must rest completely on workbenches or similar support structures due to the structural configuration of common portable router tables. As such, woodworker must sacrifice a portion of their workbench or table to accommodate such apparatus or must purchase additional devices to use such router table.

(4) In another example, woodworkers must have more than one router on hand when using a conventional router table. In these operations, the portable router that is engaged with the router table is normally disposed below the table, secured directly to the bench of the router table, and placed at a desired cutting depth. With this configuration, woodworkers have at least another router on hand when cutting operations cannot be performed by the portable router that is equipped to the portable table; such instances occur when the workpiece or stock is too large or cumbersome for the portable table. As such, woodworkers may be required to purchase various routers or similar cutting tools in order to avoid an expenditure of more time and effort when the woodworker is performing various cutting operations.

SUMMARY OF THE INVENTION

(5) In one aspect, an exemplary embodiment of the present disclosure may provide a portable router table. The portable router table includes a top surface of a bench that is adapted to support at least one workpiece, a bottom surface of the bench that is spaced apart from the top surface, and at least one bottom slot defined in the bottom surface of the bench; wherein the bench is adapted to engage with a bench clamp inside of the at least one bottom slot for clamping the portable router table with a support structure.

(6) This exemplary embodiment or another exemplary embodiment may further include that the bench is cantilevered from the support structure and the bottom surface of the bench is engaged with the support structure. This exemplary embodiment or another exemplary embodiment may further include that the bench further comprises: a front end; a rear end opposite to the front end; wherein the at least bottom slot is accessible at one of the front end and the rear end. This exemplary embodiment or another exemplary embodiment may further include that the at least one bottom slot comprises: a first bottom slot defined in the bottom surface of the bench; and a second bottom slot defined in the bottom surface of the bench and is opposite to the first bottom slot; wherein the bench is adapted to engage with the bench clamp inside of the first bottom slot for clamping the portable router table with the support structure at a first side of the bench; and wherein the bench is adapted to engage with a second bench clamp inside of the second bottom slot for clamping the portable router table with the support structure at a second side of the bench. This exemplary embodiment or another exemplary embodiment may further include a pair of legs operably engaging with the bench at the bottom surface; wherein each leg of the pair of legs is moveable between a stored position and an operating position. This exemplary embodiment or another exemplary embodiment may further include that each leg of the pair of legs comprises: a first end pivotably engaged with the bench at the bottom surface; a second end opposite to the first end and adapted to engage with the support structure; and a foot extending outwardly from the second end; wherein the bench clamp is adapted to clamp the foot with the support structure to maintain the portable router table with the support structure. This exemplary embodiment or another exemplary embodiment may further include a router base removably engaged with the bench between a first cutting orientation and a second cutting orientation; wherein when the router base is in the first cutting orientation, the router base is engaged with the bench; and wherein when the router base is in the second cutting orientation, the router base is spaced apart and remote from the bench. This exemplary embodiment or another exemplary embodiment may further include a seating recess defined in the bench and is accessible from the top surface and the bottom surface; wherein the router base is seated inside of the seating recess when in the first cutting orientation. This exemplary embodiment or another exemplary embodiment may further include an internal lateral wall; and an internal base wall positioned below the internal lateral wall; wherein the seating recess is collectively defined by the internal lateral wall and the internal base wall; and wherein the

router base is held by the internal lateral wall and the internal base wall inside of the seating recess when in the first cutting orientation. This exemplary embodiment or another exemplary embodiment may further include a front opening defined in the front end of the bench; wherein the front opening provides access to the seating recess and a handle of the router base when the router base is provided in the first cutting orientation. This exemplary embodiment or another exemplary embodiment may further include a retaining member operably engaged with the bench; wherein the retaining member is partially seated inside of the seating recess to frictionally fit the router base with the bench. This exemplary embodiment or another exemplary embodiment may further include a cavity defined by the internal lateral wall and the internal base wall and is in communication with the seating recess; wherein the retaining member is partially seated inside of the cavity. This exemplary embodiment or another exemplary embodiment may further include that the retaining member is formed of a resilient material. This exemplary embodiment or another exemplary embodiment may further include at least one top slot defined in the top surface of the bench; wherein the bench is adapted to engage with at least one clamp of a fence inside of the at least one top slot for clamping the fence with the bench.

(7) In another aspect, an exemplary embodiment of the present disclosure may provide a method of cutting a workpiece on a portable router table. The method comprises steps of: engaging a bench clamp into a slot defined in a bench of the portable router table; engaging the bench clamp and a first end of the bench, collectively, with a support structure; securing the bench clamp with the bench and the support structure; suspending a second end of the bench from the support structure, wherein the second end of the bench is opposite to the first end of the bench; and routing the workpiece on the portable router table.

(8) This exemplary embodiment or another exemplary embodiment may further include steps of resting a portion of a bottom surface of the bench on the support structure; and positioning the bench above the support structure. This exemplary embodiment or another exemplary embodiment may further include steps of resting a portion of a top surface of the bench on the support structure; and positioning the bench below the support structure such that a top surface of the bench contacts the support structure. This exemplary embodiment or another exemplary embodiment may further include steps of engaging a second bench clamp into a second slot defined in the bench, wherein the second slot is transversely opposite to the slot; engaging the second bench clamp and the first end of the bench, collectively, with the support structure; and securing the second bench clamp with the bench and the support structure. This exemplary embodiment or another exemplary embodiment may further include steps of disengaging the bench clamp from the bench and the support structure; engaging the bench clamp to a foot of a leg of the portable router table; engaging the bench clamp and the foot, collectively, with the support structure; and securing the bench clamp with the foot and the support structure. This exemplary embodiment or another exemplary embodiment may further include a step of positioning a router base of the portable router table at a first cutting orientation or at a second cutting orientation relative to the bench; wherein when the router base is in the first cutting orientation, the router base is engaged with the bench; and wherein when the router base is in the second cutting orientation, the router base is spaced apart and remote from the bench.

(9) In another aspect, an exemplary embodiment of the present disclosure may provide a portable router table. The portable router table includes a top surface of a bench that is adapted to support at least one workpiece; a bottom surface of the bench that is spaced apart from the top surface; and a router base removably engageable with the bench and is adapted to support a router; wherein when the router base is provided in a first cutting orientation, the router base is engaged with the bench; and wherein when the router base is provided in a second cutting orientation, the router base is spaced apart and remote from the bench.

(10) This exemplary embodiment or another exemplary embodiment may further include that when the router base is provided in the first cutting orientation, the router base and the top surface of the

bench are even with one another to form a continuous surface; and wherein when the router base is provided in the second cutting orientation, the router base and the top surface of the bench are free from forming a continuous surface. This exemplary embodiment or another exemplary embodiment may further include that the router base comprises: a plate removably engaged with the bench and being adapted to support the router; and a handle operably engaged with the plate. This exemplary embodiment or another exemplary embodiment may further include that the plate comprises: a first surface adapted to rest on the at least one workpiece; a second surface opposite to the first surface and is adapted to contact the router and the handle; and a central opening defined in the plate and extending between the first surface and the second surface; wherein a router bit of the router is adapted to pass through the central opening. This exemplary embodiment or another exemplary embodiment may further include that when the router base is provided in the first cutting orientation, the first surface of the plate and the top surface of the bench are even with one another to form a continuous surface; and wherein when the router base is provided in the first cutting orientation, the second surface rests inside of the bench. This exemplary embodiment or another exemplary embodiment may further include that the plate further comprises: at least two sets of apertures defined in the plate and being spaced apart from the central opening; wherein the router is adapted to engage with the router base at a first set of apertures of the at least two sets of apertures; and wherein a second router is adapted to engage with the router base at a second set of apertures of the at least two sets of apertures. This exemplary embodiment or another exemplary embodiment may further include that when the router base is provided in the first cutting orientation, the handle of the router base extends downwardly and accessible below the bottom surface of the bench. This exemplary embodiment or another exemplary embodiment may further include a seating recess defined in the bench and is accessible from the top surface and the bottom surface; wherein the router base is seated inside of the seating recess when in the first cutting orientation. This exemplary embodiment or another exemplary embodiment may further include an internal lateral wall; an internal base wall positioned below the internal lateral wall; and wherein the seating recess is collectively defined by the internal lateral wall and the internal base wall; wherein the router base is held by the internal lateral wall and the internal base wall inside of the seating recess when in the first cutting orientation. This exemplary embodiment or another exemplary embodiment may further include a front opening defined in the bench; wherein the front opening provides access to the seating recess and a handle of the router base when the router base is provided in the first cutting orientation. This exemplary embodiment or another exemplary embodiment may further include a retaining member operably engaged with the bench; wherein the retaining member is partially seated inside of the seating recess to frictionally fit the router base with the bench. This exemplary embodiment or another exemplary embodiment may further include a cavity defined by the internal lateral wall and the internal base wall and is in communication with the seating recess; wherein the retaining member is partially seated inside of the cavity. This exemplary embodiment or another exemplary embodiment may further include that the retaining member is formed of a resilient material. This exemplary embodiment or another exemplary embodiment may further include that when the router base is provided in the first cutting orientation, the router bit extends upwardly from the top surface of the bench; and wherein when the router base is provided in the first cutting orientation, the motor is positioned below the base and the at least one workpiece. This exemplary embodiment or another exemplary embodiment may further include that when the router base is provided in the first cutting orientation, a motor of the router is positioned above the at least one workpiece and remote from the bench.

(11) In another aspect, an exemplary embodiment of the present disclosure may provide a method of cutting a workpiece. The method comprises steps of: engaging a router with a router base of a portable router table; positioning the router base at a first cutting orientation or at a second cutting orientation relative to a bench of the portable router table of the portable router table; supporting a workpiece along a first surface of the router base only or along the first surface of the router and

the bench of the portable router table, collectively; and cutting a workpiece with the router.

(12) This exemplary embodiment or another exemplary embodiment may further include steps of engaging the router base with the bench of the portable router table when the router base is in the first cutting orientation; and supporting the workpiece by the router base and the bench of the portable router table when the router base is in the first cutting orientation. This exemplary embodiment or another exemplary embodiment may further include steps of separating the router base from the bench of the portable router table when the router base is in the second cutting orientation; and supporting the workpiece by the router base when the router base is in the second cutting orientation. This exemplary embodiment or another exemplary embodiment may further include steps of accessing a handle of the router base through a front opening defined in the bench of the router base; and removing the router base from the bench, by the handle, from the first cutting orientation to the second cutting orientation. This exemplary embodiment or another exemplary embodiment may further include a step of seating the router base inside of a seating recess defined in the bench when the router base is in the first cutting orientation. This exemplary embodiment or another exemplary embodiment may further include a step of securing the router base with a retaining member of the portable router table in the first cutting orientation. This exemplary embodiment or another exemplary embodiment may further include steps of disengaging the router from the router base, wherein the router is arranged with a first set of apertures defined in the router base; aligning a second router with a second set of apertures defined in the router base, wherein the second router is different than the router, and wherein the second set of apertures is different than the first set of apertures; and engaging the second router with the router base.

(13) In another aspect, an exemplary embodiment of the present disclosure may provide a portable router table. The portable router table also includes a bench; a fence operably engageable with the bench and configured to contact a workpiece; and a micro-adjuster operably engageable with the bench and the fence; wherein the micro-adjuster is configured to pivot the fence relative to a router bit of a router that is engaged with the bench.

(14) This exemplary embodiment or another exemplary embodiment may further include that when the fence is pivoted in a first pivot direction relative to the router bit, the fence is moving towards the router bit; and wherein when the fence is pivoted in a second pivot direction relative to the router bit, the fence is moving away from the router bit. This exemplary embodiment or another exemplary embodiment may further include that the fence is positioned between the bench and the micro-adjuster. This exemplary embodiment or another exemplary embodiment may further include a frame of the fence; a base plate of the micro-adjuster; a pivot mechanism of the micro-adjuster pivotably engaging the frame with the bench and the base plate; and an adjustment mechanism of the micro-adjuster moving the frame relative to the base plate. This exemplary embodiment or another exemplary embodiment may further include a first end of the fence; a second end of the fence opposite to the first end; and a measurement indicia of the micro-adjuster defined on the base plate; wherein the second end of the fence aligns with the measurement indicia to set a desired distance between the fence and the router bit. This exemplary embodiment or another exemplary embodiment may further include a starting marker that positions the fence at a first distance from the router bit; a first terminal marker that positions the fence at a second distance from the router bit; and a second terminal marker that positions the fence at a third distance from the router bit. This exemplary embodiment or another exemplary embodiment may further include that the third distance is greater than the first and second distances; and wherein the first distance is greater than the first distance. This exemplary embodiment or another exemplary embodiment may further include that the adjustment mechanism comprises: a post operably engaged with the frame; a shaft rotatably engaged with the post and the base plate; and an adjustment knob operably engaged with the shaft; wherein the adjustment knob is configured to linearly move the shaft and the post to pivot the frame of the fence. This exemplary embodiment or

another exemplary embodiment may further include that when the adjustment knob rotates in a first rotational direction, the shaft and the post move in a first linear direction to pivot the fence in a first pivot direction towards the router bit; and wherein when the adjustment knob rotates in a second rotational direction, the shaft and the post move in a second linear direction to pivot the fence in a second pivot direction away from the router bit. This exemplary embodiment or another exemplary embodiment may further include that the adjustment mechanism further comprises: a first end of the adjustment knob facing the post; a second end of the adjustment knob opposite to the first end and facing a vertical wall of the base plate; an extension of the adjustment knob extending from the second end; and a washer positioned between the adjustment knob and the vertical wall and configured to receive the extension inside a countersunk opening defined by the washer. This exemplary embodiment or another exemplary embodiment may further include that the extension is defined by a first shape and the countersunk opening of the coned washer is defined by a second shape that is complimentary with the first shape. This exemplary embodiment or another exemplary embodiment may further include that the adjustment mechanism further comprises: a spring positioned circumferentially about a portion of the shaft; and a lock nut operably engaged with the shaft wherein the spring is positioned between a vertical wall of the base plate and the lock nut to bias the shaft to a starting marker of the micro-adjuster. This exemplary embodiment or another exemplary embodiment may further include that the adjustment mechanism further comprises: a locking mechanism releasably engaged with the frame of the fence with the base plate of the micro-adjuster.

(15) In another aspect, an exemplary embodiment of the present disclosure may provide a method for adjusting a workpiece relative to a router. The method includes steps of: engaging a micro-adjuster of a portable router table to a bench of the portable router table; engaging a fence of the portable router table with the micro-adjuster and the bench; resting the workpiece along the bench and the fence; and pivoting the fence, by the micro-adjuster, in a first pivot direction or in a second pivot direction relative to a router bit until the workpiece is at a desired distance from the router bit.

(16) This exemplary embodiment or another exemplary embodiment may further include steps of engaging a post of the micro-adjuster with the fence; engaging a shaft of the micro-adjuster with the fence and a base plate of the micro-adjuster; engaging an adjustment knob of the micro-adjuster with the shaft and the base plate. This exemplary embodiment or another exemplary embodiment may further include that the step of pivoting the fence further comprises: rotating an adjustment knob of the micro-adjuster a first rotational direction or in a second rotational direction; moving the shaft in a first linear direction or in a second linear direction; moving the post and the fence, collectively, in the first pivot direction towards the router bit or in the second pivot direction away from the router bit. This exemplary embodiment or another exemplary embodiment may further include a step of positioning the micro-adjuster between the fence and the bench. This exemplary embodiment or another exemplary embodiment may further include steps of aligning a rear end of a frame of the fence with a starting marker of a measurement indicia that is on a base plate of the micro-adjuster; and defining the fence at a first distance measured relative to the router bit. This exemplary embodiment or another exemplary embodiment may further include steps of aligning the rear end of the frame with a first terminal marker of the measurement indicia; defining the fence at a second distance measured relative to the router bit that is less than the first distance. This exemplary embodiment or another exemplary embodiment may further include steps of aligning the rear end of the frame with a second terminal marker of the measurement indicia; and defining the fence at a third distance measured relative to the router bit that is greater than the first distance and the second distance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Sample embodiments of the present disclosure are set forth in the following description, are shown in the drawings and are particularly and distinctly pointed out and set forth in the appended claims.
- (2) FIG. 1 (FIG. 1) is a front, top, first side isometric perspective view of a portable router table and a removable router base in accordance with one aspect of the present disclosure, wherein the removable router base is in the first cutting orientation.
- (3) FIG. 2 (FIG. 2) is a rear, top, first side isometric perspective view of the portable router table and the removable router base in accordance with one aspect of the present disclosure.
- (4) FIG. 3 (FIG. 3) is a partial top plan view of the portable router table and the removable router base.
- (5) FIG. 4 (FIG. 4) is a sectional view of the portable router base taken in the direction of line 4-4 shown in FIG. 3.
- (6) FIG. 5 (FIG. 5) is a partial rear elevation of the portable router base taken in the direction of line 5-5 shown in FIG. 2.
- (7) FIG. 6 (FIG. 6) is a partial bottom, rear, second side perspective view of a bottom slot defined in the portable router table.
- (8) FIG. 7 (FIG. 7) is an operational view of the removable router base being removed from the portable router table at the first cutting orientation.
- (9) FIG. 8 (FIG. 8) is an exploded view of the removable router base.
- (10) FIG. 9 (FIG. 9) is a rear, bottom, first side isometric perspective view of the portable router table, wherein a pair of legs of the portable router base is in an extended position.
- (11) FIG. 10 (FIG. 10) is a rear, bottom, first side isometric perspective view of the portable router table, wherein the pair of legs of the portable router base is translated from the extended position to a collapsed position.
- (12) FIG. 11 (FIG. 11) is a top perspective view of an exemplary clamp.
- (13) FIG. 12 (FIG. 12) is an operational view of the portable router base supported to a support structure, wherein clamps are secured to feet of the pair of legs and the removable router base is in the first cutting orientation.
- (14) FIG. 13A (FIG. 13A) is an operational view of clamps operably engaging with a bench of the portable router base.
- (15) FIG. 13B (FIG. 13B) is another operational view continuing from FIG. 13A, wherein a handle of the clamps are shown being moveable for clamping operations.
- (16) FIG. 13C (FIG. 13C) is another operational view continuing from FIG. 13B, wherein the portable router table is engaged to the support structure by the clamps, and wherein the portable router table is cantilevered from the support structure and the removable router base is in the first cutting orientation.
- (17) FIG. 14A (FIG. 14A) is an operational view of the removable router base and router being removed from the portable router table.
- (18) FIG. 14B (FIG. 14B) is another operational view continuing from FIG. 14A, wherein the removable router base is rotated to the second cutting orientation.
- (19) FIG. 14C (FIG. 14C) is another operational view continuing from FIG. 14A, wherein the removable router base is in the second cutting orientation and routes a workpiece remote from the portable router base.
- (20) FIG. 15 (FIG. 15) is a partial rear, top, second side isometric perspective view of the portable router table in accordance with another aspect of the present disclosure, wherein the portable router table includes an alternative fence and a micro-adjuster assembly.
- (21) FIG. 16 (FIG. 16) is a partial top plan view of the portable table shown in FIG. 15, wherein the fence is positioned at a starting point and is measured at a first distance from a router bit equipped

to the router

(22) FIG. 16A (FIG. 16A) is an enlargement of the highlighted region shown in in FIG. 16.

(23) FIG. 17 (FIG. 17) is a sectional view of a portion of the fence and micro-adjuster assembly shown in FIG. 15.

(24) FIG. 18A (FIG. 18A) is an operational view of the portable router table, wherein the micro-adjuster assembly translates the fence in a first direction from the starting position to a first terminal position and measures at a second distance from the router bit.

(25) FIG. 18B (FIG. 18B) is an operational view of the portable router table, wherein the micro-adjuster assembly translates the fence in a second direction from the starting position to a second terminal position and measures at a third distance from the router bit.

(26) FIG. 19 (FIG. 19) is an exemplary method flowchart.

(27) FIG. 20 (FIG. 20) is an exemplary method flowchart.

(28) FIG. 21 (FIG. 21) is an exemplary method flowchart.

(29) Similar numbers refer to similar parts throughout the drawings.

DETAILED DESCRIPTION

(30) FIGS. 1-2 illustrates a portable router table 1 that assists a router in routing and/or cutting various types of workpiece. As discussed in greater detail below, portable router table 1 is selectively operably engageable with a support structure in a first cutting configuration and in a second cutting configuration for assisting in routing and/or cutting various types of workpiece. Such components of the portable router table 1 are now discussed in greater detail below.

(31) Portable router table 1 includes a bench 10. As best seen in FIG. 1, bench 10 includes a first or front end 10a, a second or rear end 10b opposite to the first end 10a, and a transverse axis defined between the first end 10a and the second end 10b. Bench 10 also includes a first side 10c, a second side 10d opposite to the first side 10c, and a longitudinal axis defined between the first side 10c and the second side 10d. Bench 10 also includes a top surface 10e, a bottom surface 10f opposite to the top surface 10e, and a vertical axis defined between the top surface 10e and the bottom surface 10f. In the present disclosure, bench 10 also defines a two-dimensional rectangular shape when view from atop perspective view (see FIGS. 1-2). In other exemplary embodiments, bench 10 may define any suitable two-dimensional shape when viewed from a top plan view as dictated by the implementation of bench 10.

(32) Still referring to bench 10, bench 10 also defines a front opening 10g. As best seen in FIGS. 1 and 7, front opening 10g extends transversely into the bench 10 from the first end 10a towards the second end 10b. The top surface 10e and the bottom surface 10f of the bench 10 are also in communication with one another at the front opening 10g. In the present disclosure, the front opening 10g defines an arcuate and/or curvilinear shape when viewed from a top plan view (see FIG. 3). As such, a portion of the first end 10a matches the arcuate and/or curvilinear shape in order to define the front opening 10g in the bench 10. In other exemplary embodiments, front opening 10g defined in bench 10 may have any suitable shape as dictated by the implementation of bench 10. Such use and purpose of front opening 10g is discussed in greater detail below.

(33) Still referring to bench 10, bench 10 also defines a seating recess 10h. As best seen in FIG. 7, seating recess 10h is collectively defined by an internal lateral wall 10i that extends vertically downward into the front opening 10g from the top surface 10e towards the bottom surface 10f and terminates at an internal base wall 10j. Such use and purpose of the seating recess 10h is discussed in greater detail below.

(34) Bench 10 also defines a cavity 10k that extends vertically downward into the bench 10 from the top surface 10e. Particularly, cavity 10k is defined by a second internal lateral wall 10i' and terminates at a second internal base wall 10j'. In the present disclosure, the cavity 10k is also in communication with the seating recess 10h; such communication between the seating recess 10h and the cavity 10k is discussed in greater detail below. It should be noted that second internal lateral wall 10i' and second internal base wall 10j' are continuous with the internal lateral wall 10i

and the internal base wall **10j**.

(35) Bench **10** also defines a through-hole **10m** that extends vertically between the bottom surface **10f** and the internal base wall **10j** (see FIG. 4). In the present disclosure, the bottom surface **10f** and the internal base wall **10j** are in communication with one another at the through-hole **10m**. Such use and purpose of through-hole **10m** is discussed in greater detail below.

(36) Still referring to bench **10**, bench **10** defines a first or top pair of slots **10n**. As best seen in FIGS. 1-2, each slot of the top pair of slots **10n** is defined in the top surface **10e** of bench **10** between the front opening **10g** and a respective side **10c**, **10d** of bench **10**. Each slot of the top pair of slots **10n** is also defined proximate to the second end **10b** of the bench **10** such that each slot of the top pair of slots **10n** extends transversely from the second end **10b** towards the first end **10a**. Each slot of the top pair of slots **10n** is spaced apart from and/or is remote from the first end **10a** of the bench **10**. As discussed in greater detail below, bench **10** is configured to receive and engage with a fence discussed herein at one or both of the slots of the top pair of slots **10n** for assistance in routing or cutting various types of workpieces.

(37) Still referring to bench **10**, bench **10** defines a second or bottom pair of slots **10p**. As best seen in FIGS. 1-2, each slot of the bottom pair of slots **10p** is defined in the bottom surface **10f** of bench **10** between the front opening **10g** and a respective side **10c**, **10d** of bench **10**. Each slot of the bottom pair of slots **10p** is also defined proximate to the second end **10b** of the bench **10** such that each slot of the bottom pair of slots **10p** extends transversely from the second end **10b** towards the first end **10a**. As such, each slot of the bottom pair of slots **10p** is spaced apart from and/or is remote from the first end **10a** of the bench **10**. As discussed in greater detail below, clamps or similar attachment devices may engage with the bench **10** at one or both of the slots of the bottom pair of slots **10p** to clamp the bench **10** to a support structure in which the bench **10** is cantilevered from said support structure.

(38) In the present disclosure, each slot of the top pair of slots **10n** and the bottom pair of slots **10p** has a T-shaped cross-section when viewed from a rear elevation view (see FIG. 5). Such T-shaped configuration of each slot of the top pair of slots **10n** and the bottom pair of slots **10p** may allow a F-clamp or similar clamp to engage with the bench **10**. In other exemplary embodiments, each slot of the top pair of slots **10n** and the bottom pair of slots **10p** may define any suitable shape as dictated by the implementation of bench **10**, including the type of clamp that may be used with the bench **10**.

(39) Portable router table **1** also includes a first pair of mounts **12** that operably engage with the bench **10**. As best seen in FIG. 9, each mount of the first pair of mounts **12** operably engages with the bottom surface **10f** of the bench **10** and extends away from the bench **10**. Still referring to FIG. 9, each mount of the first pair of mounts **12** is also positioned proximate to the first side **10c** of the bench **10** and is positioned between a first slot of the bottom pair of slots **10p** and the front opening **10g**. Similarly, portable router table **1** also includes a second pair of mounts **14** that operably engage with the bench **10**; the second pair of mounts **14** is positioned longitudinally opposite to the first pair of mounts **12** on bench **10** relative to the longitudinal axis of bench **10**. As best seen in FIG. 9, each mount of the second pair of mounts **14** operably engages with the bottom surface **10f** of the bench **10** and extends away from the bench **10**. Still referring to FIG. 9, each mount of the second pair of mounts **14** is also positioned proximate to the second side **10d** of the bench **10** and is positioned between a second slot of the bottom pair of slots **10p** and the front opening **10g**. Such use and purpose of the first pair of mounts **12** and the second pair of mounts **14** are discussed in greater detail below.

(40) Portable router table **1** also includes a pair of legs **20a**, **20b**. As best seen in FIG. 9, each leg of the pair of legs **20a**, **20b** operably engages with the bench **10** at the bottom surface **10f**. Particularly, each leg of the pair of legs **20a**, **20b** pivotably engages with the first pair of mounts **12** and the second pair of mounts **14**. As discussed in greater detail below, each leg of the pair of legs **20a**, **20b** may transition between an extended or operating position (see FIG. 9) and a collapsed or stored

position (see FIG. 10) depending on the intended use of portable router table 1. Such collapsing of the pair of legs 20a, 20b from the operating position to the stored position is denoted by arrows labeled “C” in FIG. 10.

(41) It should be understood that each leg of the pair of legs 20a, 20b is identical to one another, but the first leg 20a of the pair of legs 20 pivotably engages with the first pair of mounts 12 and the second leg 20b of the pair of legs 20 pivotably engages with the second pair of mounts 14. For brevity, the first leg 20a of the pair of legs 20 will be discussed herein given the identical configuration with the second leg 20b of the pair of legs 20. It should be understood that while the first leg 20a of the pair of legs 20 is discussed herein, such components and/or features apply equally to the second leg 20b of the pair of legs 20.

(42) As best seen in FIG. 9, first leg 20a includes a first or top end 22a and a second or bottom end 22b that is opposite to the first end 22a. First leg 20a also includes a foot 22c that extends outwardly from a bend 22d defined in first leg 20a to the second end 22b of first leg 20a; as such, foot 22c includes the second end 22b. In the present disclosure, second end 22b and foot 22c are collectively offset from the first end 22a relative to a lengthwise axis the extends between the first end 22a to the bend 22d. As discussed in greater detail below, foot 22c is configured to engage with a work surface or support surface of a support structure to provide stability in a first plane to the portable router table 1 when the first leg 20a is in the operating position.

(43) Still referring to FIG. 9, first leg 20a also includes a front support 22d that extends lengthwise between the first end 22a and the second end 22b of first leg 20a and faces in the same direction as the first end 10a of bench 10. First leg 20a also includes a rear support 22e that extends lengthwise between the first end 22a and the second end 22b of first leg 20a and faces in the same direction as the second end 10b of bench 10. In the present disclosure, the front support 22d and the rear support 22e face in opposing directions relative to one another. The front support 22d and the rear support 22e are also offset from the foot 22c relative to the lengthwise axis of first leg 20a due to the front support 22e and the rear support 22f extending in a second direction from a first vertical bend 22g and a second vertical bend 22h. In the present disclosure, each of the first vertical bend 22g and the second vertical bend 22h extends lengthwise between the first end 22a and horizontal bend 22d along the lengthwise axis of first leg 20a. As discussed in greater detail below, the front support 22e and the rear support 22f are configured to engage with bench 10 and a work surface or support surface of a support structure to provide stability in a second plane to the portable router table 1 when the first leg 20a is in the operating position; the second plane is substantially parallel with the first plane of the foot 22c.

(44) Still referring to first leg 20a, first leg 20a also defines a pair of pivot holes 22i. As best seen in FIG. 9, a first pivot hole of the pair of pivot holes 22i is defined in the front support 22e of first leg 20a, and a second pivot hole of the pair of pivot holes 22i is defined in the rear support 22f of first leg 20a. As discussed in greater detail below, the pair of pivot holes 22i of first leg 20a is aligned with openings in the first pair of mounts 12 so that pivot mechanisms of portable router table 1 pivotably engage first leg 20a with the bench 10. Similarly, such alignment is identical between the second pair of mounts 14 and second leg 20b so that pivot mechanisms of portable router table 1 pivotably engage second leg 20b with the bench 10.

(45) Still referring to first leg 20a, foot 22c defines a set of attachment holes 22j. As best seen in FIG. 9, each attachment hole of the set of attachment holes 22j extends entirely through foot 22c in which a connector, fastener, or bolt may pass through foot 22c for fixedly engaging the foot 22c with a support structure. Such use of the set of attachment holes 22j may be desired by a woodworker when the woodworker intends to keep or maintain the portable router table 1 permanently or to provide a more rigid configuration to prevent any shifting of the table 1 when routing or cutting various types of workpiece.

(46) Still referring to first leg 20a, first leg 20a also defines a set of cross members or struts 22k between the first end 22a and the second end 22b. As best seen in FIG. 9, each cross member 22k

provides stability to first leg **20a** in both a lateral or transverse direction as well as in a vertical direction. Such inclusion of cross members **22k** also eliminates weight from first leg **20a** by removing material from first leg **20a** instead of having a solid wall extending between the first end **22a** and the second end **22b**. It should be understood, however, that in other exemplary embodiments, such legs discussed herein may include solid walls if desired.

(47) Portable router table **1** also includes a set of pivot mechanisms **30** that pivotably engage the pair of legs **20a**, **20b** with the first pair of mounts **12** and the second pair of mounts **14**. As best seen in FIG. **9**, two pivot mechanisms of the set of pivot mechanism **30** pivotably engage first leg **20a** with the first pair of mounts **12** so that first leg **20a** is pivotable between the operating position (see FIG. **9**) and the stored position (FIG. **10**). Similarly, two different pivot mechanisms of the set of pivot mechanism **30** pivotably engage second leg **20b** with the second pair of mounts **14** so that second leg **20b** is pivotable between the operating position (see FIG. **9**) and the stored position (FIG. **10**). It should be understood that each pivot mechanism of the set of pivot mechanisms **30** may include any suitable components in order for the pair of legs **20a**, **20b** to be pivotable between the operating position (see FIG. **9**) and the stored position (FIG. **10**). In the present disclosure, each pivot mechanism of the set of pivot mechanisms **30** includes a connector that pivotably engages a respective leg **20a**, **20b** with a respective mount **12**, **14** discussed herein and is maintained with the respective leg **20a**, **20b** and the respective mount **12**, **14** by a nut or retaining member. Each pivot mechanism of the set of pivot mechanisms **30** also includes a pivot axis **30a** that extends along the length of the connector of each pivot mechanism of the set of pivot mechanism **30**; the respective leg **20a**, **20b** pivots about the pivot axis **30a** when moving from the operating position to the stored position.

(48) Portable router table **1** also includes a removable router base **40** that is selectively operably engageable with the bench **10**. As discussed in greater detail below, removable router base **40** (hereinafter “base **40**”) may be moveable between a first cutting orientation (see FIGS. **1-3** and FIGS. **12** and **13C**) to a second cutting orientation (see FIGS. **7** and FIGS. **14A-14C**) depending on the type of workpiece that must be routed and/or cut with a router attached to the removable router base **40**. Such components of the base **40** are discussed in greater detail below.

(49) Base **40** includes a plate **42** that is removably engageable with the bench **10** inside of the front opening **10g** and the seating recess **10h**. As best seen in FIG. **8**, plate **42** includes a front or first end **42a**, a rear or second end **42b** that is opposite to the first end **42a**, and a lengthwise axis defined therebetween. In the present disclosure, the second end **42b** defines a rounded edge that corresponds to the rounded and/or curvilinear shape of the front opening **10g** collectively defined by the internal lateral wall **10i** and the internal base wall **10j**. Still referring to FIG. **8**, plate **42** also includes a first surface **42c** that extends lengthwise between the first end **42a** and the second end **42b** and faces in a first direction. Plate **42** also includes a second surface **42d** that extends lengthwise between the first end **42a** and the second end **42b** and faces in a second direction that is opposite to the first direction of first surface **42c**.

(50) Still referring to plate **42**, plate **42** also defines a router bit opening **42e**. As best seen in FIG. **8**, router bit opening **42e** is defined between the first end **42a** and the second end **42b** but is closer to the second end **42b**. Router bit opening **42e** also extends entirely through plate **42** between the first surface **42c** and the second surface **42d** such that the first surface **42c** and the second surface **42d** are in communication with one another at the router bit opening **42e**. As discussed in greater detail below, router bit opening **42e** is configured to allow a router bit that is equipped to a router to pass through the plate **42** when said router is operably engaged with the plate **42**.

(51) Still referring to plate **42**, plate **42** also defines a countersunk aperture **42f**. As best seen in FIG. **8**, countersunk aperture **42f** is defined between the first end **42a** and the router bit opening **42e** but is closer to the first end **42b**. Countersunk aperture **42f** also extends entirely through plate **42** between the first surface **42c** and the second surface **42d** such that the first surface **42c** and the second surface **42d** are in communication with one another at the countersunk aperture **42f**. As

discussed in greater detail below, countersunk aperture **42f** enables a handle of base **40** to operably engage with the plate **42** so that a woodworker may easily grip and control base **40** during routing operations.

(52) Still referring to plate **42**, plate **42** also defines a plurality of attachment apertures **42g**. As best seen in FIG. **8**, the plurality of attachment apertures **42g** is defined circumferentially about the router bit opening **42e** close to the second end **42b**. Each attachment aperture of the plurality of attachment apertures **42g** also extends entirely through plate **42** between the first surface **42c** and the second surface **42d** such that the first surface **42c** and the second surface **42d** are in communication with one another at each attachment aperture of the plurality of attachment apertures **42g**. It should be understood that the plurality of attachment apertures **42g** includes individual sets of attachment apertures that enables a woodworker to operably engage a commercially-available router to the plate **42** based on the respective set of attachment apertures of the plurality of attachment apertures.

(53) Still referring to plate **42**, plate **42** also includes a peripheral wall **42h**. As best seen in FIG. **8**, peripheral wall **42** extends continuously between the first surface **42c** and the second surface **42d**. As discussed in greater detail below, peripheral wall **42h** may contact and/or interface with the internal lateral wall **10i** of bench **10** when the base **40** is in the first cutting orientation. As also discussed in greater detail below, peripheral wall **42h** may also contact and/or interface with a retaining member of router table **1** to maintain the base **40** with the bench **1** in the first cutting orientation.

(54) As mentioned previously, base **40** also includes a handle **44** that is operably engaged with the plate **42** by a bolt **46**. As best seen in FIG. **1**, handle **44** extends downwardly from the plate **42** when the handle **44** is engaged with the plate **42** by the bolt **46** at the countersunk aperture **42f**. Here, the handle **44** is directly contacting the second surface **42d** and is positioned below the first surface **42c** so that the handle **44** does not impede or obstruct the woodworker when routing or cutting various types of workpiece along the bench **10** when the base **40** is provided in a first cutting orientation (see FIGS. **1-2**, **12**, and **13C**). Additionally, the head of the bolt **46** is even with or positioned below the first surface **42c** of the plate **42** to prevent marring or damage to a workpiece as said workpiece passes over the first surface **42c** of plate **42** when the woodworker is routing or cutting various types of workpiece along the bench **10** in the first cutting orientation.

(55) As briefly mentioned above, base **40** is selectively engageable with the bench **10** between a first cutting orientation (see FIGS. **1-2**, **12**, and **13C**) and a second cutting orientation (see FIGS. **14A-14C**). In the first cutting orientation, base **40** is housed inside of the front opening **10g** such that the top surface **10e** of bench **10** and the first surface **42c** of plate **42** of base **40** are even with one another to collectively define a continuous flat or planar surface. Such continuous surface allows a woodworker to move a workpiece along the bench **10** and base **40** with ease without being impeded or obstructed during a routing or cutting operation. In this first cutting orientation, base **40** is also housed and held inside of seating recess **10h** collectively by the internal lateral wall **10i** and the internal base wall **10j**. Particularly, the plate **42** of base **40** is housed inside of seating recess **10h** and is collectively held by the internal lateral wall **10i** and the internal base wall **10j** inside of the front opening **10g** at the second surface **42d**.

(56) Still referring to the first cutting orientation, handle **44** of base **40** is also positioned below the bench **10**. As discussed previously, such positioning of the handle **44** below the bench **10** in the first cutting orientation prevents the woodworker from being impeded or obstructed from moving various types of workpiece along the bench **10** during routing or cutting operations. In the first cutting orientation, the handle **44** is also easily accessible to a woodworker since the woodworker may simply pass his/her hand through the front opening **10g** at the first end **10a** of the bench **10** for grasping or gripping said handle **44**. Such ease of accessibility enables a woodworker to transition the base **40** from the first cutting orientation to the second cutting orientation when workpiece is too large or cumbersome to cut or route on the bench **10**.

(57) In the second cutting orientation, base **40** is completely removed and spaced apart from bench **10** (see FIGS. **14A-14C**). To accomplish this second cutting orientation, woodworker passes his/her hand through the front opening **10g** at the first end **10a** of the bench **10** and grasps the handle **44**. As illustrated in FIG. **14A**, woodworker applies an upward forward on the base **40**, by the handle **44**, until the base **40** and the attached router (discussed below) are removed from the front opening **10g** of bench **10**. At this stage, woodworker may then rotate base and the attached router (FIG. **14B**) to route or cut various types of workpiece that are too large or cumbersome for bench **10**.

(58) Portable router table **1** also includes a retaining mechanism **50** that operably engages with the bench **10**. As best seen in FIG. **4**, retaining mechanism **50** includes a retaining member **51** that is substantially housed inside of cavity **10k** of bench **10** while a portion of the retaining member **51** projects into the front opening **10g**. Retaining mechanism **50** also includes a bolt **52** that threadably engages with a nut **54** to attach the retaining member **51** with the bench **10** at the through-hole **10m**.

(59) It should be understood that retaining member **51** is made from a resilient and/or elastic material that is able to deform when a force is applied against the retaining member **51**. As best seen in FIG. **3**, the portion of the retaining member **51** that projects into the front opening **10g** deforms when the plate **42** of base **40** directly contacts the retaining member **51**, particularly the peripheral wall **42h** of plate **42**. When such contact occurs between the plate **42** and retaining member **51**, retaining member **51** applies a retention force on the plate **42** so that the plate **42** in frictional fit between at least the internal lateral wall **10i** and the retaining member **51**. Such retention force applied on the plate **42** prevents the plate **42** from disengaging the bench **10** when the portable router table **1** is transported, when a router tool is engaged with the plate **42**, or when the portable router table **1** is being used to route or cut various types of workpiece. It should be understood that such retention force may be overcome by the woodworker when the woodworker applies an upward force on the base **40** at the handle **44**.

(60) Portable router table **1** also includes a fence **60** that is selectively operably engageable with the bench **10**. As best seen in FIGS. **1-2**, fence **60** includes a frame **62** that is selectively operably engageable with the bench **10** inside of the top pair of slots **10n** by a pair of clamping handles **64**. Fence **60** also includes a guide **66** that is fixedly engaged with the frame **62**. In operation, woodworker may move fence **60** transversely between first end **10a** and second end **10b** to set a desired cutting distance between the guide **66** and a router bit equipped to a router.

(61) Having now discussed the components of the portable router table **1**, methods of using the portable router table **1** at various orientations and/or configurations for cutting operations are discussed in greater detail below.

(62) Prior to engaging the portable router table **1** with a support structure or similar woodworking tool, woodworker may engage a router **80** with the plate **42** of base **40** at the plurality of attachment apertures **42g**. As stated previously, a group or set of attachment apertures of the plurality of attachment apertures **42g** is selected by the woodworker based on the manufacturer of said router **80**. Once the set of apertures of the plurality of attachment apertures **42g** are selected, the router **80** is engaged with the plate **42** via a set of fasteners **82** (see FIG. **3**). A router bit **84** is also equipped to the router **80** based on the desired routing or cutting operations that woodworker is about to perform. As best seen in FIGS. **1-2** and **12**, router bit **84** projects through the router bit opening **42e** defined in the plate **42** of base **40** where the plate **42** is free from interfering with the router bit **84** when rotated by router **80**.

(63) FIG. **12** illustrates a first configuration of portable router table **1** where portable router table **1** is placed and clamped to a support structure **90** where the portable router table **1** is entirely on top of the support structure **90**. In the present embodiment, at least one clamp **100** engages with each leg of the pair of legs **20a**, **20b** to maintain the portable router table **1** with support structure **90**. Particularly, at least one clamp **100** engages with the foot **22c** of each leg of the pair of legs **20a**, **20b** to maintain the portable router table **1** with support structure **90**. In this example, support structure **90** is a router table where the support structure **90** includes a table **92** that defines a

plurality of clamp holes **94** for receiving clamps **100**.

(64) Such first configuration may be desirable for a woodworker when the woodworker is not limited on space along the table **92** of support structure **90** or needs additional height when routing or cutting a workpiece with portable router table **1**.

(65) FIGS. **13A-13C** illustrates a second configuration of portable router table **1** where portable router table **1** is clamped to support structure **90** where the portable router table **1** is suspended and/or cantilevered from the support structure **90**. Prior to suspension of portable router table **1**, woodworker must first introduce and slide a fixed jaw **100a** of each clamp **100** into a respective slot of the bottom pair of slots **10p** defined in the bench **10**; such sliding of the clamps **100** into the bench is denoted by arrows labeled “M1” in FIG. **13A**.

(66) In this configuration, the woodworker slides the fixed jaw **100a** of each clamp **100** into the second end **10b** of the bench **10** until the fixed jaw **100a** of each clamp **100** is seated inside of the respective slot of the bottom pair of slots **10p**. Once seated, the woodworker may then collectively move the portable router table **1** and the clamps **100** towards an edge **92a** of the table **92** until the fixed jaw **100a** of each clamp **100** is substantially contacting the top surface of the table **92** for clamping means. The woodworker may then translate a moveable jaw **100b** of each clamp **100** towards the bottom surface of table **92** and the respective fixed jaw **100a** until the clamp **100** provides adequate clamping force to cantilever the portable router table **1** from the support structure **90**; such translation of moveable jaws **100b** is denoted by an arrow labeled “M2” in FIG. **13B**. Prior to clamping the moveable jaw **100b** of each clamp **100** to the table **92**, the woodworker slides the portable router table **1** and the clamps **100**, collectively, onto the table **92** at a respective edge or end of the table (e.g., front edge **92a**); such sliding of the portable router table **1** and the clamps **100**, collectively, onto the table **92** is denoted by an arrow labeled “M3” in FIG. **13C**.

(67) Such second configuration may be desirable for a woodworker when the woodworker is limited on space along the table **92** of support structure **90** or needs the entire surface area of the table **92** for other woodworking needs along with routing and/or cutting various types of workpiece. As such, the woodworker does not need to sacrifice or dedicate a section of the table **92** when a separate woodworking operation is needed on table **92**. In one example, woodworker may perform a first woodworking operation when portable router table **1** is cantilevered from a support structure **90** (e.g., routing or cutting various types of workpiece with router **80** when base **40** is in the first orientation) and a second woodworking operation on table **92** that is separate from the first woodworking operation. Such second woodworking operation may be any suitable woodworking operation that is commonly performed by woodworkers or is commonly performed by a woodworker when using router **80** or other woodworking tools.

(68) FIGS. **14A-14C** illustrate a third configuration of portable router table **1** where base **40** is removed from bench **10** for routing or cutting a stock or workpiece that is too large or cumbersome to cut along bench **10**. As such, base **40** is oriented from the first orientation (see FIGS. **1** and **12**) to the second orientation (see FIG. **14A**) in which the base **40** is removed and separated from the bench **10** by the woodworker. Such removal of the base **40** by the woodworker is denoted by an arrow labeled “R1” in FIG. **14A**.

(69) Once removed, woodworker may then rotate the base **40** and the router **80** 180 degrees from the first orientation to the second orientation. As best seen in FIG. **14B**, the base **40** and router **80** have been rotated to the second orientation by the woodworker in which the router **80** is now positioned above and/or extends vertically from the second surface **42d** of base **40**. Such rotation of base **40** and router **80** from the first orientation to the second orientation is denoted by arrows labeled “R2” in FIG. **14B**.

(70) Once in the second orientation, the woodworker may then collectively introduce the base **40** and the router **80** to a desired workpiece (workpiece denoted as “WP” in FIG. **14C**). Once the base **40** is resting on the workpiece, the woodworker may then operate the router **80** to route or cut into the workpiece for various reasons. As discussed previously, base **40** acts a planar or flat support so

that router **80** is free from swaying or titling as the woodworker runs the router **80** along a portion of the workpiece. As best seen in FIG. **14C**, base **40** and router **80** are placed on workpiece in which router **80** cuts into a straight edge of workpiece “WP1” to create a rounded edge “WP2” along workpiece.

(71) FIGS. **15-18B** illustrate a micro-adjuster assembly **120** that operably engages with an alternative frame **62'** of fence **60'**. It should be noted that frame **62'** is similar to frame **62** of fence **60** discussed above and illustrated in FIGS. **1-2**, except as detailed below. It should also be noted that clamping handles **64'** and guide **66'** of fence **60'** are identical to the clamping handles **64** and guide **66** discussed above. The following components and features of the frame **62'** and micro-adjuster assembly **120** are now discussed in greater detail below.

(72) As best seen FIG. **15**, frame **62'** includes a first end **62a'** that operably engages with guide **66'**, a second end **62b'** that is opposite to the first end **62a'**, and a first axis defined therebetween. Frame **62'** also includes a first side **62c'**, a second side **62d'** that is opposite to the first side **62c'**, and a second axis defined therebetween which is orthogonal to the first axis. Frame **62'** also includes an outer surface **62e'** that contacts guide **66'** and base plate of micro-adjuster assembly **120**, and an inner surface **62f'** that is opposite to the outer surface **62e'** and is free from contacting guide **66'**.

(73) Still referring to FIG. **15**, frame **62'** defines a T-shaped configuration when viewed from a side elevation view. In the present disclosure, frame **62'** includes a first major wall **62A'** that operably engages with guide **66'**, and a second major wall **62B'** that operably engages with a base plate of a micro-adjuster assembly. Additionally, frame **62'** defines a bend **62C'** that separates the first major wall **62A'** and the second major wall **62B'** from one another. With such structural configuration, the first major wall **62A'** and the second major wall **62B'** define a continuous, unitary member from the first end **62a** to the second end **62b**. In other exemplary embodiments, first and second major walls of a frame may be two separate piece that are attached to one another to collectively define the frame.

(74) Still referring to frame **62'**, frame **62'** also defines a pivot opening **62g'** and a locking slot **62h'**. As best seen in FIG. **16**, pivot opening **62g'** extends entirely through frame **62'** between the outer surface **62e'** and the inner surface **62f'**; the outer surface **62e'** and the inner surface **62f'** are in communication with one another at the pivot opening **62g'**. It should be noted that pivot opening **62g'** is also closer to the second end **62b'** of frame **62'**. As best seen in FIG. **16A**, locking slot **62h'** also extends entirely through frame **62'** between the outer surface **62e'** and the inner surface **62f'**; the outer surface **62e'** and the inner surface **62f'** are in communication with one another at the locking slot **62h'**. It should be noted that locking slot **62h'** is also closer to the second end **62b'** of frame **62'** and opposite to the pivot opening **62g'** relative to the second axis of the frame **62'**. Such use and purpose of pivot opening **62g'** and locking slot **62h'** are discussed in greater detail below.

(75) Micro-adjuster assembly **120** includes a base plate **122**. As best seen FIG. **15**, base plate **122** includes a first end **122a**, a second end **122b** that is opposite to the first end **62a'**, and a first axis defined therebetween. Base plate **122** also includes a first side **122c**, a second side **122d** that is opposite to the first side **122c**, and a second axis defined therebetween which is orthogonal to the first axis of base plate **122**. Base plate **122** also includes a top surface **122e**, an bottom surface **122f** that is opposite to the top surface **122e**, and a third axis defined therebetween which is orthogonal to the first axis of base plate **122**.

(76) Still referring to base plate **122**, base plate **122** also includes a pivot post **122g** and a locking post **122h**. As best seen in FIG. **16**, pivot post **122g** extends upwardly from the top surface **122e** of the base plate **122**. It should be noted that pivot post **122g** is also closer to the second end **122b** of base plate **122**. As best seen in FIG. **16A**, locking post **122h** also extends upwardly from the top surface **122e** of the base plate **122**. It should be noted that locking post **122h** is also closer to the second end **122b** of base plate **122** and opposite to the pivot post **122g** relative to the second axis of the base plate **122**. Upon assembly with frame **62'**, the pivot post **122g** extends through the pivot opening **62g'** of the frame **62'**, and locking post **122h** extends through the locking slot **62h'** of frame

62'. Such use and purpose of the pivot post **122g** and locking post **122h** are discussed in greater detail below.

(77) Base plate **122** also defines a slot **122i**. As best seen in FIG. **16A**, slot **122i** extends entirely through base plate **122** between the top surface **122e** and the bottom surface **122f**; the top surface **122e** and the bottom surface **122f** are in communication with one another at the slot **122i**. It should be noted that slot **122i** is also closer to the second end **122b** of base plate **122** and opposite to the pivot post **122g** relative to the second axis of the base plate **122**. In the present disclosure, the locking post **122h** is defined between the pivot post **122g** and the slot **122i** relative to the second axis of base plate **122**. Such use and purpose of the slot **122i** are discussed in greater detail below.

(78) Still referring to base plate **122**, base plate **122** may define a front notch. In this embodiment, front notch may extend inwardly from the first end **122a** towards the second end **122b** and is defined between the first side **122c** and the second side **122d**. The front notch may be configured to receive and house a portion of a tubular member of a vacuum system that is engaged with the frame **62'** of the fence **60'**. In the present disclosure embodiment, front notch is omitted from base plate **122** when a tubular member of a vacuum system is omitted from the fence **60'**.

(79) Still referring to base plate **122**, base plate **122** also includes a vertical wall **122k**. As best seen in FIGS. **15** and **17**, vertical wall **122k** is bent upwardly from an internal wall of base plate **122** and is positioned above the top surface **122e**. It should be noted that vertical wall **122k** includes a portion of the second end **122b** of base plate **122** that is positioned vertically above the top surface **122e** and the remaining second end **122b** of base plate **122**. It should also be noted that vertical wall **122k** includes a portion of top surface **122e** and a portion of the bottom surface **122f** that is positioned vertically above remaining top surface **122e** and bottom surface **122f**. Due to such construction of base plate **122**, base plate **122** also defines a rear cutout **122m** in the second end **122b** of the base plate **122**. Still referring to vertical wall **122k**, vertical wall **122k** also defines an aperture **122n** that extends entirely through the vertical wall **122k** between the top surface **122e** and the bottom surface **122f** (see FIG. **17**); the top surface **122e** and the bottom surface **122f** are in communication with one another at the aperture **122n**. Such use and purpose of vertical wall **122k** and aperture **122n** are discussed in greater detail below.

(80) Micro-adjuster assembly **120** also includes a pivot mechanism **124** that operably engages frame **62'** and base plate **122** with one another. As best seen in FIGS. **15-16**, pivot mechanism **124** pivotably engages the frame **62'** and the base plate **122** with one another at the pivot opening **62g'** of the frame **62'** when the pivot mechanism **124** threadably engages with pivot post **122g** of the base plate **122**. Particularly, a nut washer **124a** of pivot mechanism **124** threadably engages with the pivot post **122g** to pivotably engage the frame **62'** and the base **122** with one another. Pivot mechanism **124** also includes a nylon washer **124b** that is positioned between the nut washer **124a** and the second major wall **62B'** of frame **62'** to allow a woodworker to easily pivot the frame **62'**. Upon adjustment, the frame **62'** pivots about a pivot axis **124c** defined along the length of the pivot post **122** for setting fence **60'** to different cutting distance relative to the router bit **84** and/or the base plate **122**. Such adjustment of frame **62'** with micro-adjuster assembly **120** is discussed in further detail below.

(81) Micro-adjuster assembly **120** also includes a post **126** that operably engages with frame **62'**. As best seen in FIG. **17**, post **126** operably engages with the frame **62'** at the pair of post openings **62i'** by a pair of fasteners **127** and is positioned above the slot **122i** of base plate **122**. Post **126** also defines a threaded passageway **126a** that extends along an axis that is perpendicular to the length of post **126**; such use and purpose of threaded passageway **126a** is discussed in greater detail below. It should be noted that post **126** is fixed to the frame **62'** such that the post **126** is stationary during adjustment operations performed on the micro-adjuster assembly by a woodworker.

(82) Micro-adjuster assembly **120** also includes a threaded shaft **128** that threadably engages with post **126**. As best seen in FIG. **17**, threaded shaft **128** includes a first end **128a** that threadably engages with the threaded passageway **126a** of post **126**, a second end **128b** that is longitudinally

opposite to the first end **128a** and is spaced apart from the post **126**, and a rotational axis **128c** extending between the first end **128a** and the second end **128b**. The first end **128a** of the threaded shaft **128** is also maintained inside of the post **126** by a set screw **130** that threadably engages with the threaded passageway **126a** of post **126** and directly abuts the first end **128a** of threaded shaft **128** (see FIG. 17). In the present disclosure, the set screw **130** acts a stop to prevent the threaded shaft **128** from extending entirely through the post **126** when the threaded shaft **128** is adjusted by a woodworker, which is discussed in greater detail below.

(83) Micro-adjuster assembly **120** also includes an adjustment knob **132** that threadably engages with the threaded shaft **128**. As best seen in FIG. 17, adjustment knob **132** includes a first end **132a** that faces the post **126**, a second end **132b** that is opposite to the first end **132a** and faces the vertical wall **122k** of base plate **122**, and a lengthwise axis extending between the first end **132a** and the second end **132b**. Adjustment knob **132** also defines a threaded passage **132c** that extends entirely through the adjustment knob **132** along the length of said adjustment knob **132** between the first end **132a** and the second end **132b**. In operation, and as discussed in greater detail below, the threaded passage **132c** enables the threaded shaft **128** to threadably engage with the adjustment knob **132** in order to linear move the adjustment knob **132** to adjust the position of the frame **62'** relative to the router bit **84** and/or the base plate **122**. Adjustment knob **132** may also defining a grip **132d** that is positioned external to the threaded passage **132c** and extends circumferentially about the adjustment knob **132**. Adjustment knob **132** also includes an extension **132e** that extends longitudinally from the second end **132b**. As best seen in FIG. 17, the extension **132e** also defines an engagement surface **132e'**. In the present disclosure, the engagement surface **132e'** has a curvilinear or rounded shape when viewed from a sectional view (see FIG. 17); such use of a curvilinear or rounded cross-sectional shape to define extension **132e** is discussed in greater detail below. In other exemplary embodiments, an engagement surface of an extension mentioned herein may define any suitable shape based on the implementation of extension **132e**.

(84) Micro-adjuster assembly **120** also includes a concave washer **134**. As best seen in FIG. 17, concave washer **134** is installed onto the threaded shaft **128** and is positioned between the vertical wall **122k** and the adjustment knob **132**, particularly the extension **132e**. Upon assembly, a portion of the extension **132e** operably engages inside of a countersunk opening **134a** defined in the concave washer **134**. The shape of the countersunk opening **134a** defined in the concave washer **134** is curvilinear and/or rounded; as such, the shapes of the extension **132e** and the countersunk opening **134a** of the concave washer **134** match one another. Such complementary shapes allows the adjustment knob **132** to pivot inside of the concave washer **134** when the fence **60'** pivots based on adjustments made by the woodworker when using the micro-adjuster assembly **120**.

(85) Micro-adjuster assembly **120** also includes a spring **136**. As best seen in FIG. 17, the spring **136** operably engages with the vertical wall **122k** of the base plate **122** and with a lock nut **138**. In the present disclosure, the spring **136** surrounds a portion of the threaded shaft **128** that is positioned behind the vertical wall **122k** and the lock nut **138**. The lock nut **138** is also threadably engaged with the threaded shaft **128** at the end that opposes the end threadably engaged with the post **126**. Once assembled, the lock nut **138** acts as a retaining mechanism or stop in order for the spring **136** to continuously apply a biasing force on the base plate **122**. Such biasing force applied by the spring **36** prevents the woodworker from manually pushing or pulling the base plate **122** when adjusting the frame **62'** as well as removes the backlash when adjusting the frame **62'** with the micro-adjuster assembly **120**.

(86) Micro-adjuster assembly **120** may also include a pair of flat washers **140a**, **140b**. As best seen in FIG. 17, a first flat washer **140a** is positioned between the vertical wall **122k** and the first end **136a** of spring **136** to prevent the spring **136** from entering into the aperture **122n** defined in the vertical wall **122k** during adjustment operations. Additionally, a second flat washer **140b** is positioned between the vertical wall **122k** and the first end **136a** of spring **136** to prevent the spring **136** from entering into the aperture **122n** defined in the vertical wall **122k** during adjustment

operations

(87) Micro-adjuster assembly **120** also includes a locking mechanism **142** that releasably engages the frame **62'** and the base plate **122** with one another at a location that spaced apart from the pivot mechanism **124**. As best seen in FIG. **16A**, locking mechanism **142** includes a locking handle **142a** that threadably engages with the locking post **122h** which passes through the locking slot **62h'** of the frame **62'**. Upon assembly, locking handle **142a** releasably locks the frame **62'** and the base plate **122** with one another. As discussed in greater detail below, the woodworker may tighten the locking handle **142a** with the locking post **122h** to clamp and secure the frame **62'** and the base plate **122** with one another to maintain the frame **62'** at a desired angle relative to the base plate **122** for cutting operation.

(88) Micro-adjuster assembly **120** also includes measurement indicia **144**. As best seen in FIG. **16A**, the measurement indicia **144** is located on the base plate **122**, particularly the top surface **122e** of the base plate **122**. The measurement indicia **144** includes a center of travel marker or starting marker **144a** that indicates that the second end **62b'** of the frame **62'** is at a first or starting distance (denoted "D1" in FIG. **16**) that is measured relative to the base plate **122** or the router bit **84**. The measurement indicia **144** also includes a first terminal marker **144b** and a second terminal marker **144c** opposite to the first terminal marker **144b**. With respect to the first terminal marker **144b**, marker **144b** indicates that the fence **60'** is set at a first terminal distance (denoted "D2" in FIG. **18A**) that is measured relative to the base plate **122** or the router bit **84**. Similarly, with respect to the second terminal marker **144c**, marker **144c** indicates that the fence **60'** is set at a second terminal distance that is measured relative to the base plate **122** or the router bit **84** (denoted as "D3" in FIG. **18B**). In operation, the woodworker uses the rearmost end or edge of the fence **60'** (i.e., second end **62b'**) to set the desired distance at one of the starting marker **144a**, the first terminal marker **144b**, or the second terminal marker **144c**.

(89) It should be noted that the woodworker may also set the second end **62b'** of fence **60'** at other suitable locations based on the cutting or routing operations. In one example, woodworker may set the second end **62b'** of the frame **62'** at a location between the starting marker **144a** and the first terminal marker **144b**. In another example, woodworker may set the second end **62b'** of the frame **62'** at a location between the starting marker **144a** and the second terminal marker **144c**. While not illustrated herein, smaller or incremental markers may be placed between the starting marker **144a** and the first terminal marker **144b** to provide more precise and accurate fine adjustments when the woodworker is adjusting the frame **62'** in the first adjustment direction. Similarly, smaller or incremental markers may be placed between the starting marker **144a** and the second terminal marker **144c** to provide more precise and accurate adjustments when the woodworker is adjusting the frame **62'** in the second adjustment direction.

(90) As mentioned previously, the first terminal marker **144b** and the second terminal marker **144c** each set the frame **62'** to a specific distance measured relative to the starting marker **144a**. In one example, the first terminal distance measured between the starting marker **144a** and the first terminal marker **144b** is approximately 0.22 inches. In the same example, the second terminal distance measured between the starting marker **144a** and the second terminal marker **144c** is approximately 0.22 inches. With this versatility, a woodworker may perform fine adjustments in a first direction (i.e., moving second end **62b'** of fence **60'** from the starting marker **144a** to the first terminal marker **144b**) where the woodworker wants incrementally move the fence **60'** closer to the router bit **84**. Similarly, a woodworker may also perform fine adjustments in a second opposing direction (i.e., moving second end **62b'** of fence **60'** from the starting marker **144a** to the second terminal marker **144c**) where the woodworker wants incrementally move the fence **60'** away from the router bit **84**.

(91) Having now discussed the components and features of the frame **62'** and the micro-adjuster assembly **120**, a method of setting a desired cutting distance between the fence **60'** and the router bit **84** with the micro-adjuster assembly **120** is discussed in greater detail below.

(92) Prior to setting a desired cutting distance between the fence **60'** and the router bit **84** with the micro-adjuster assembly **120**, the woodworker releasably secures the fence **60'** and the micro-adjuster assembly **120** with the bench **10** by the clamping handles **64**. Particularly, the woodworker may collectively slide the fence **60'** and the micro-adjuster assembly **120** along the bench **10** until the fence **60'** and the micro-adjuster assembly **120** reaches a desired location on the bench **10**. Such desired location may be based on the size, shape, or configuration of the various types of workpiece that the woodworker is cutting while using the fence **60'** and the micro-adjuster assembly **120**. Once the desired location is reached, the clamping handles **64** are tightened to the base plate **122** (at the clamping openings **122p**) and the bench **10** (at the top pair of slots **10n**) until the base plate **122** is secured to the bench **10**.

(93) Once secured to the bench **10**, the woodworker may then use the micro-adjuster assembly **120** to set the desired cutting distance for the frame **62'**. Prior to setting such cutting distance, the frame **62'** is set at the starting marker **144a** based on the placement of the second end **62b'** of the frame **62'**. With this configuration, frame **62'** is set at the first distance "D1" measured between the router bit **84** when referencing to the starting marker **144a**.

(94) To initiate adjustment, the woodworker rotates the adjustment knob **132** in either a first rotational direction or a second rotational direction to move the fence **60'**; such rotation of the adjustment knob **132** is denoted by an arrow labeled "RT1" in FIG. **18A**. For illustrative purposes, the adjustment knob **132** is being rotated in the first rotational direction in FIG. **18A** to show the functionality of the micro-adjuster assembly **120**. The rotation of the adjustment knob **132** in the second rotational direction is substantially similar but performed in the opposing direction; such rotation of the adjustment knob **132** in the second rotational direction is denoted by an arrow labeled "RT2" in FIG. **18B**.

(95) As the adjustment knob **132** is rotated in the first rotational direction, the threaded shaft **128** is rotated by the adjustment knob **132** causing the threaded shaft **128** to linearly move in a first linear direction. With such linear movement, the threaded shaft **128** presses against the post **126** in the first direction causing the frame **62'** to move in the first direction due to the engagement between the frame **62'** and the post **126**. As the frame **62'** is being moved by the threaded shaft **128**, the frame **62'** also pivots about the connector **124a** of pivot mechanism **124** at a distance away from the post **126** for setting the desired cutting distance for woodworker. As the frame **62'** is being moved by the threaded shaft **128**, the spring **136** also compresses due to the threaded shaft **128** being moved linearly away from the vertical wall **122k**.

(96) As the frame **62'** is pivoting about the connector **124a**, the threaded shaft **128** also moves inside of the aperture **122n** defined in the vertical wall **122k**. As such, the aperture **122n** is wide enough so that the vertical wall **122k** is free from impeding or hindering movement of the threaded shaft **128** as the threaded shaft **128** pivots with the frame **62'**. It should also be noted that the interaction between the extension **132e** and the concave washer **134** enables the adjustment knob **132** to pivot on the concave washer **134** with ease due to the matching shapes of the extension **132e** and the rounded and/or concave opening of the concave washer **134**.

(97) Once the woodworker reaches the desired cutting distance, the woodworker ceases actuation of the adjustment knob **132** to cease all movement between the frame **62'** and the micro-adjuster assembly **120**. Such actuation of the adjustment knob **132** may conclude when the second end **62b'** of the frame **62** reaches a desired marker, such as the first terminal marker **144b** or a marker located between the starting marker **144a** and the first terminal marker **144b**, that matches with the desired cutting distance. Once the desired cutting distance is reached, the woodworker may then tighten the locking mechanism **142** to the frame **62'** and the base plate **122** to maintain the desired cutting distance. Particularly, the woodworker screws or tightens the locking handle **142a** with the locking post **122h** until the frame **62'** and the base plate **122** are releasably secured with one another and are free from shifting or sliding away from the desired cutting distance. At this stage, the woodworker may then introduce a workpiece to router table **1** once the fine adjustments are complete.

(98) In the present disclosure, the first distance “D1”, the second distance “D2”, the third distance “D3” provide the fence 60' at different distance relative to the router bit 84. In the present disclosure, when the second end 62b' of fence 60' is aligned with the first terminal marker 144b (see FIG. 18A), the second distance “D2” is set and is less than the first distance “D1” and the third distance “D3”. Additionally, when the second end 62b' of fence 60' is aligned with the second terminal marker 144c (see FIG. 18B), the third distance “D3” is greater than the first distance “D1”.

(99) It should be noted that woodworker may repeat the adjustment process one or more times depending on the types of workpiece that the woodworker will be cutting with the router table 1.

(100) FIG. 19 is a method 200 of cutting a workpiece on a portable router table. An initial step 202 of method 200 includes engaging a bench clamp into a slot defined in a bench of the portable router table. Another step 204 of method 200 includes engaging the bench clamp and a first end of the bench, collectively, with a support structure. Another step 206 of method 200 includes securing the bench clamp with the bench and the support structure. Another step 208 of method 200 includes suspending a second end of the bench from the support structure, wherein the second end of the bench is opposite to the first end of the bench. Another step 210 of method 200 includes routing the workpiece on the portable router table.

(101) In other exemplary embodiments, additional or optional steps may be included in method 200 for cutting a workpiece with a portable router table. In one exemplary embodiment, method 200 may further include steps of resting a portion of a bottom surface of the bench on the support structure; and positioning the bench above the support structure. In another exemplary embodiment, method 200 may further include steps of resting a portion of a top surface of the bench on the support structure; and positioning the bench below the support structure. In another exemplary embodiment, method 200 may further include steps of engaging a second bench clamp into a second slot defined in the bench, wherein the second slot is transversely opposite to the slot; engaging the second bench clamp and the first end of the bench, collectively, with the support structure; and securing the second bench clamp with the bench and the support structure. In another exemplary embodiment, method 200 may further include steps of disengaging the bench clamp from the bench and the support structure; engaging the bench clamp to a foot of a leg of the portable router table; engaging the bench clamp and the foot, collectively, with the support structure; and securing the bench clamp with the foot and the support structure. In another exemplary embodiment, method 200 may further include steps of positioning a router base of the portable router table at a first cutting orientation or at a second cutting orientation relative to the bench; wherein when the router base is in the first cutting orientation, the router base is engaged with the bench; and wherein when the router base is in the second cutting orientation, the router base is spaced apart and remote from the bench.

(102) FIG. 20 is a method 300 of cutting a workpiece. An initial step 302 of method 300 includes engaging a router with a router base of a portable router table. Another step 304 of method 300 includes positioning the router base at a first cutting orientation or at a second cutting orientation relative to a bench of the portable router table of the portable router table. Another step 306 of method 300 includes supporting a workpiece along a first surface of the router base only or along the first surface of the router and the bench of the portable router table, collectively. Another step 308 of method 300 includes cutting a workpiece with the router.

(103) In other exemplary embodiments, additional or optional steps may be included in method 300 for cutting a workpiece with a portable router table. In one exemplary embodiment, method 300 may further include steps of engaging the router base with the bench of the portable router table when the router base is in the first cutting orientation; and supporting the workpiece by the router base and the bench of the portable router table when the router base is in the first cutting orientation. In another exemplary embodiment, method 300 may further include steps of separating the router base from the bench of the portable router table when the router base is in the second

cutting orientation; and supporting the workpiece by the router base when the router base is in the second cutting orientation. In another exemplary embodiment, method **300** may further include steps of accessing a handle of the router base through a front opening defined in the bench of the router base; and removing the router base from the bench, by the handle, from the first cutting orientation to the second cutting orientation. In another exemplary embodiment, method **300** may further include a step of seating the router base inside of a seating recess defined in the bench when the router base is in the first cutting orientation. In another exemplary embodiment, method **300** may further include a step of securing the router base with a retaining member of the portable router table in the first cutting orientation. In another exemplary embodiment, method **300** may further include steps of disengaging the router from the router base, wherein the router is arranged with a first set of apertures defined in the router base; aligning a second router with a second set of apertures defined in the router base, wherein the second router is different than the router, and wherein the second set of apertures is different than the first set of apertures; and engaging the second router with the router base.

(104) FIG. **21** is a method **400** for adjusting a workpiece relative to a router. An initial step **402** of method **400** includes engaging a micro-adjuster of a portable router table to a bench of the portable router table. Another step **404** of method **400** includes engaging a fence of the portable router table with the micro-adjuster and the bench. Another step **406** of method **400** includes resting the workpiece along the bench and the fence. Another step **408** of method **400** includes pivoting the fence, by the micro-adjuster, in a first pivot direction or in a second pivot direction relative to a router bit until the workpiece is at a desired distance from the router bit.

(105) In other exemplary embodiments, additional or optional steps may be included in method **400** for cutting a workpiece with a portable router table. In one exemplary embodiment, method **400** may further include steps of engaging a post of the micro-adjuster with the fence; engaging a shaft of the micro-adjuster with the fence and a base plate of the micro-adjuster; engaging an adjustment knob of the micro-adjuster with the shaft and the base plate. In another exemplary embodiment, method **400** may further include that the step of pivoting the fence further comprises: rotating an adjustment knob of the micro-adjuster a first rotational direction or in a second rotational direction; moving the shaft in a first linear direction or in a second linear direction; moving the post and the fence, collectively, in the first pivot direction towards the router bit or in the second pivot direction away from the router bit. In another exemplary embodiment, method **400** may further include a step of positioning the micro-adjuster between the fence and the bench. In another exemplary embodiment, method **400** may further include steps of aligning a rear end of a frame of the fence with a starting marker of a measurement indicia that is on a base plate of the micro-adjuster; and defining the fence at a first distance measured relative to the router bit. In another exemplary embodiment, method **400** may further include steps of aligning the rear end of the frame with a first terminal marker of the measurement indicia; defining the fence at a second distance measured relative to the router bit that is less than the first distance. In another exemplary embodiment, method **400** may further include steps of aligning the rear end of the frame with a second terminal marker of the measurement indicia; and defining the fence at a third distance measured relative to the router bit that is greater than the first distance and the second distance.

(106) Various inventive concepts may be embodied as one or more methods, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

(107) Any flowchart and/or block diagrams in the Figures illustrate some exemplary architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments of the present disclosure. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of

instructions, which comprises one or more executable instructions for implementing the specified logical function(s). In some alternative implementations, the functions noted in the blocks may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions

(108) While various inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

(109) The articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” The phrase “and/or,” as used herein in the specification and in the claims (if at all), should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc. As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

(110) As used herein in the specification and in the claims, the phrase “at least one,” in reference to

a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

(111) While components of the present disclosure are described herein in relation to each other, it is possible for one of the components disclosed herein to include inventive subject matter, if claimed alone or used alone. In keeping with the above example, if the disclosed embodiments teach the features of A and B, then there may be inventive subject matter in the combination of A and B, A alone, or B alone, unless otherwise stated herein.

(112) As used herein in the specification and in the claims, the term “effecting” or a phrase or claim element beginning with the term “effecting” should be understood to mean to cause something to happen or to bring something about. For example, effecting an event to occur may be caused by actions of a first party even though a second party actually performed the event or had the event occur to the second party. Stated otherwise, effecting refers to one party giving another party the tools, objects, or resources to cause an event to occur. Thus, in this example a claim element of “effecting an event to occur” would mean that a first party is giving a second party the tools or resources needed for the second party to perform the event, however the affirmative single action is the responsibility of the first party to provide the tools or resources to cause said event to occur.

(113) When a feature or element is herein referred to as being “on” another feature or element, it can be directly on the other feature or element or intervening features and/or elements may also be present. In contrast, when a feature or element is referred to as being “directly on” another feature or element, there are no intervening features or elements present. It will also be understood that, when a feature or element is referred to as being “connected”, “attached” or “coupled” to another feature or element, it can be directly connected, attached or coupled to the other feature or element or intervening features or elements may be present. In contrast, when a feature or element is referred to as being “directly connected”, “directly attached” or “directly coupled” to another feature or element, there are no intervening features or elements present. Although described or shown with respect to one embodiment, the features and elements so described or shown can apply to other embodiments. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed “adjacent” another feature may have portions that overlap or underlie the adjacent feature.

(114) Spatially relative terms, such as “under”, “below”, “lower”, “over”, “upper”, “above”, “behind”, “in front of”, and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if a device in the figures is inverted, elements described as “under” or “beneath” other elements or features would then be oriented “over” the other elements or features. Thus, the exemplary term “under” can encompass both an orientation of over and under. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly. Similarly, the terms “upwardly”, “downwardly”, “vertical”,

“horizontal”, “lateral”, “transverse”, “longitudinal”, and the like are used herein for the purpose of explanation only unless specifically indicated otherwise.

(115) Although the terms “first” and “second” may be used herein to describe various features/elements, these features/elements should not be limited by these terms, unless the context indicates otherwise. These terms may be used to distinguish one feature/element from another feature/element. Thus, a first feature/element discussed herein could be termed a second feature/element, and similarly, a second feature/element discussed herein could be termed a first feature/element without departing from the teachings of the present disclosure.

(116) An embodiment is an implementation or example of the present disclosure. Reference in the specification to “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” “an exemplary embodiment,” or “other embodiments,” or the like, means that a particular feature, structure, or characteristic described in connection with the embodiments is included in at least some embodiments, but not necessarily all embodiments, of the invention. The various appearances “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” “an exemplary embodiment,” or “other embodiments,” or the like, are not necessarily all referring to the same embodiments. Furthermore, the use of any and all examples or exemplary language (“e.g.,” “such as,” or the like) is intended merely to better illustrate or illuminate the embodiments and does not pose a limitation on the scope of that or those embodiments. No language in this specification should be construed as indicating any unclaimed element as essential to the practice of the disclosed embodiment.

(117) If this specification states a component, feature, structure, or characteristic “may”, “might”, or “could” be included, that particular component, feature, structure, or characteristic is not required to be included. If the specification or claim refers to “a” or “an” element, that does not mean there is only one of the element. If the specification or claims refer to “an additional” element or “another” element, that does not preclude there being more than one of the additional element or the another element.

(118) As used herein in the specification and claims, including as used in the examples and unless otherwise expressly specified, all numbers may be read as if prefaced by the word “about” or “approximately,” even if the term does not expressly appear. The phrase “about” or “approximately” may be used when describing magnitude and/or position to indicate that the value and/or position described is within a reasonable expected range of values and/or positions. For example, a numeric value may have a value that is $\pm 0.1\%$ of the stated value (or range of values), $\pm 1\%$ of the stated value (or range of values), $\pm 2\%$ of the stated value (or range of values), $\pm 5\%$ of the stated value (or range of values), $\pm 10\%$ of the stated value (or range of values), etc. Any numerical range recited herein is intended to include all sub-ranges subsumed therein. Further, recitation of ranges of values herein are not intended to be limiting, referring instead individually to any and all values falling within that range, unless otherwise indicated herein, and each separate value within such range is incorporated into the specification as if it were individually recited herein.

(119) Additionally, the method of performing the present disclosure may occur in a sequence different than those described herein. Accordingly, no sequence of the method should be read as a limitation unless explicitly stated. It is recognizable that performing some of the steps of the method in a different order could achieve a similar result.

(120) In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-closed transitional phrases, respectively.

(121) To the extent that the present disclosure has utilized the term “invention” in various titles or sections of this specification, or in the context of those sections, this term has been included as

required by the formatting requirements of word document submissions (i.e., docx submissions) pursuant the guidelines/requirements of the United States Patent and Trademark Office and shall not, in any manner, be considered a disavowal of any subject matter.

(122) In the foregoing description, certain terms have been used for brevity, clearness, and understanding. No unnecessary limitations are to be implied therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed.

(123) Moreover, the description and illustration of various embodiments of the disclosure are examples and the disclosure is not limited to the exact details shown or described.

Claims

1. A method of cutting a workpiece on a portable router table, comprising: engaging a bench clamp into a slot defined in a bench of the portable router table; engaging the bench clamp and a first end of the bench, collectively, with a support structure; securing the bench clamp with the bench and the support structure; suspending a second end of the bench from the support structure, wherein the second end of the bench is opposite to the first end of the bench; routing the workpiece on the portable router table; disengaging the bench clamp from the bench and the support structure; engaging the bench clamp to a foot of a leg of the portable router table; engaging the bench clamp and the foot, collectively, with the support structure; and securing the bench clamp with the foot and the support structure.
2. The method of claim 1, wherein the step of engaging the bench clamp and the first end of the bench with the support structure comprises positioning the bench above the support structure such that a bottom surface of the bench contacts the support structure.
3. The method of claim 1, wherein the step of engaging the bench clamp and the first end of the bench with the support structure comprises positioning the bench below the support structure such that a top surface of the bench contacts the support structure.
4. The method of claim 1, further comprising: engaging a second bench clamp into a second slot defined in the bench, wherein the second slot is transversely opposite to the slot; engaging the second bench clamp and the first end of the bench, collectively, with the support structure; and securing the second bench clamp with the bench and the support structure.
5. The method of claim 1, wherein the bench is configured to be cantilevered from the support structure, and wherein a bottom surface of the bench is configured to be engaged with the support structure.
6. The method of claim 1, further comprising: defining at least one top slot in a top surface of the bench; wherein the bench is adapted to engage with at least one clamp of a fence inside of the at least one top slot for clamping the fence with the bench.
7. The method of claim 1, further comprising: providing a pair of side walls that extends between a front end of the bench and a rear end of the bench; wherein the slot is defined between a side wall of the pair of side walls and a leg of a pair of legs.
8. The method of claim 1, further comprising: providing a pair of side walls that extends between a front end of the bench and a rearend of the bench; wherein the slot is defined between a first side wall of the pair of side walls and a leg of a pair of legs; and wherein a second slot is defined between a second side wall of the pair of side walls and a second leg of the pair of legs.
9. The method of claim 1, further comprising: positioning a router base of the portable router table at a first cutting orientation or at a second cutting orientation relative to the bench; wherein when the router base is in the first cutting orientation, the router base is engaged with the bench; and wherein when the router base is in the second cutting orientation, the router base is spaced apart and remote from the bench.
10. The method of claim 9, further comprising: defining a seating recess in the bench and being

accessible from a top surface of the bench and a bottom surface of the bench; wherein the router base is seated inside of the seating recess when in the first cutting orientation.

11. The method of claim 10, further comprising: providing an internal lateral wall; and providing an internal base wall positioned below the internal lateral wall; wherein the seating recess is collectively defined by the internal lateral wall and the internal base wall; and wherein the router base is held by the internal lateral wall and the internal base wall inside of the seating recess when in the first cutting orientation.

12. The method of claim 10, further comprising: defining a front opening in the front end of the bench; and providing access to the seating recess and a handle of the router base, by the front opening, when the router base is provided in the first cutting orientation.

13. The method of claim 10, further comprising: engaging a retaining member with the bench; and partially seating the retaining member inside of the seating recess; and directly abutting a peripheral wall of the plate of the router base with the retaining member to frictionally fit the router base with the bench.

14. The method of claim 13, further comprising: defining a cavity by the internal lateral wall and the internal base wall and being in communication with the seating recess; wherein the retaining member is partially seated inside of the cavity.

15. The method of claim 13, wherein the retaining member is formed of a resilient material.
